

Row Reduction and Echelon Forms

Definitions

Zero Column

A column with *all* entries equal to zero

Zero Row

A row with *all* entries equal to zero

Nonzero Column

A column with *at least one* entry not equal to zero

Nonzero Row

A row with *at least one* entry not equal to zero

Leading Entry of a row is the leftmost nonzero entry

0 0 0 **2** 0 0 1 3
→

Row Reduction and Echelon Forms

Definition

A rectangular matrix is in **echelon form** (or **row echelon form**) if it has the following three properties:

1. All nonzero rows are above any rows of all zeros.
2. Each leading entry of a row is in a column to the right of the leading entry of the row above it.
3. All entries in a column below a leading entry are zeros.

$$\begin{bmatrix} \blacksquare & * & * & * \\ 0 & \blacksquare & * & * \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & \blacksquare & * & * & * & * & * & * & * & * \\ 0 & 0 & 0 & \blacksquare & * & * & * & * & * & * \\ 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \blacksquare & * \end{bmatrix}$$

Each matrix is row equivalent to **one and only one reduced echelon matrix**.

If a matrix in echelon form satisfies the following additional conditions, then it is in **reduced echelon form** (or **reduced row echelon form**):

4. The leading entry in each nonzero row is 1.
5. Each leading 1 is the only nonzero entry in its column.

$$\begin{bmatrix} 1 & 0 & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 & * & 0 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 1 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 1 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 1 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & * \end{bmatrix}$$

$$\begin{bmatrix} 2 & -3 & 2 & 1 \\ 0 & 1 & -4 & 8 \\ 0 & 0 & 0 & 5/2 \end{bmatrix}$$

Echelon Form

$$\begin{bmatrix} 1 & 0 & 0 & 29 \\ 0 & 1 & 0 & 16 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

Reduced Echelon Form

**Echelon Form is
not unique**

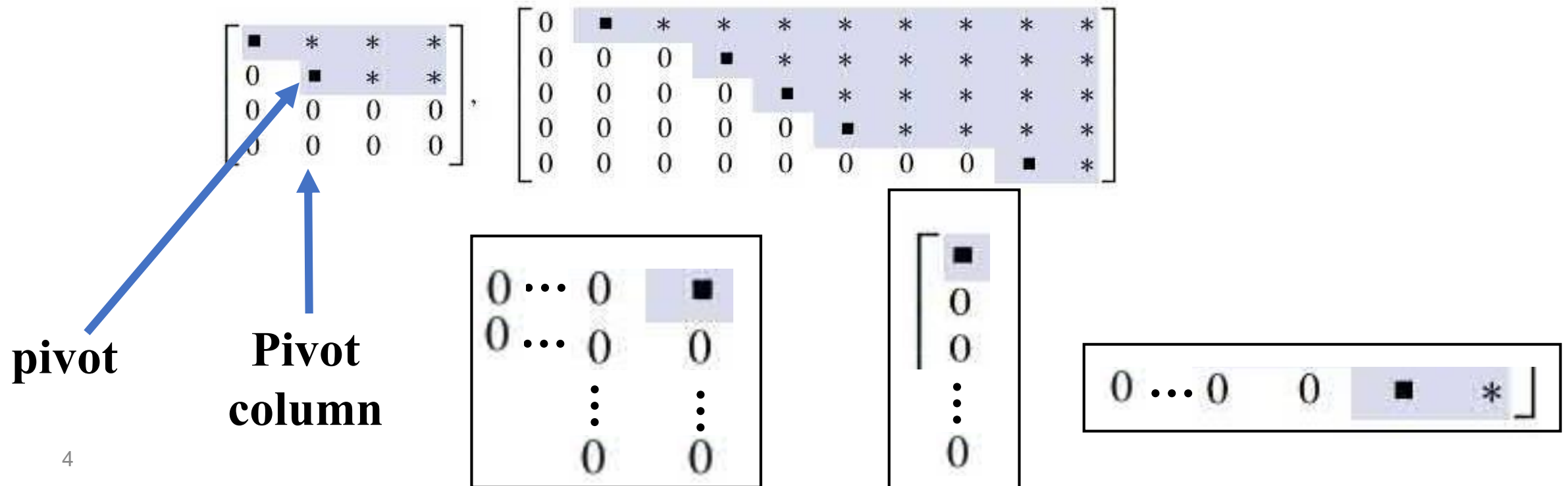
A nonzero matrix can have **more than one** equivalent matrices in **Echelon Forms**

**Reduced Echelon
Form is unique**

A nonzero matrix can have **only one** equivalent matrix in **Reduced Echelon Forms**

pivots

A **pivot position** in a matrix A is a location in A that corresponds to a leading 1 in the reduced echelon form of A . A **pivot column** is a column of A that contains a pivot position.



Algorithms to produces a matrix in **Echelon Form**

Step 1

Begin with the leftmost nonzero column. This is a pivot column. The pivot position is at the top.

Step 2

Select a nonzero entry in the pivot column as a pivot. If necessary, interchange rows to move this entry into the pivot position.

Step 3

Use row replacement operations to create zeros in all positions below the pivot.

Step 4

Cover (or ignore) the row containing the pivot position and cover all rows, if any, above it. Apply steps 1–3 to the submatrix that remains. Repeat the process until there are no more nonzero rows to modify.

Example

Row reduce the matrix A below to echelon form, and locate the pivot columns in A .

$$A = \begin{bmatrix} 0 & -3 & -6 & 4 & 9 \\ -1 & -2 & -1 & 3 & 1 \\ -2 & -3 & 0 & 3 & -1 \\ 1 & 4 & 5 & -9 & -7 \end{bmatrix}$$

interchange R1 and R4

$$\begin{bmatrix} 1 & 4 & 5 & -9 & -7 \\ -1 & -2 & -1 & 3 & 1 \\ -2 & -3 & 0 & 3 & -1 \\ 0 & -3 & -6 & 4 & 9 \end{bmatrix}$$

Create 0s here

Pivot

Pivot column

$R2 = R1 + R2$
 $R3 = R3 + 2R1$

$$\begin{bmatrix} 1 & 4 & 5 & -9 & -7 \\ 0 & 2 & 4 & -6 & -6 \\ 0 & 5 & 10 & -15 & -15 \\ 0 & -3 & -6 & 4 & 9 \end{bmatrix}$$

Create 0s here

Next pivot column

Pivot

$R3 = R3 - 5/2R2$
 $R4 = R4 + 3/2R2$

$$\begin{bmatrix} 1 & 4 & 5 & -9 & -7 \\ 0 & 2 & 4 & -6 & -6 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -5 & 0 \end{bmatrix}$$

interchange R3 and R4

$$\begin{bmatrix} 1 & 4 & 5 & -9 & -7 \\ 0 & 2 & 4 & -6 & -6 \\ 0 & 0 & 0 & -5 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Pivot columns

General form:

$$\begin{bmatrix} \blacksquare & * & * & * & * \\ 0 & \blacksquare & * & * & * \\ 0 & 0 & 0 & \blacksquare & * \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & -3 & -6 & 4 & 9 \\ -1 & -2 & -1 & 3 & 1 \\ -2 & -3 & 0 & 3 & -1 \\ 1 & 4 & 5 & -9 & -7 \end{bmatrix}$$

Pivot positions

Pivot columns

6

Pivot columns: C1, C2 and C4
Pivots: 1, 2, -5

Algorithms to produce a matrix in Reduced Echelon Form

Step 1

Begin with the leftmost nonzero column. This is a pivot column. The pivot position is at the top.

Step 2

Select a nonzero entry in the pivot column as a pivot. If necessary, interchange rows to move this entry into the pivot position.

Step 3

Use row replacement operations to create zeros in all positions below the pivot.

Step 4

Cover (or ignore) the row containing the pivot position and cover all rows, if any, above it. Apply steps 1–3 to the submatrix that remains. Repeat the process until there are no more nonzero rows to modify.

**Forward
Phase**

Same steps
of echelon
form

+

One more
step

Step 5

Beginning with the rightmost pivot and working upward and to the left, create zeros above each pivot. If a pivot is not 1, make it 1 by a scaling operation.

**Backward
Phase**

Example

Apply elementary row operations to transform the following matrix first into echelon form and then into reduced echelon form:

$$\begin{aligned} 3x_2 - 6x_3 + 6x_4 + 4x_5 &= -5 \\ 3x_1 - 7x_2 + 8x_3 - 5x_4 + 8x_5 &= 9 \\ 3x_1 - 9x_2 + 12x_3 - 9x_4 + 6x_5 &= 15 \end{aligned}$$

$$\begin{bmatrix} 0 & 3 & -6 & 6 & 4 & -5 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 3 & -9 & 12 & -9 & 6 & 15 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 3 & -6 & 6 & 4 & -5 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 3 & -9 & 12 & -9 & 6 & 15 \end{bmatrix}$$

interchange R1 and R3

$$\begin{bmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 0 & 3 & -6 & 6 & 4 & -5 \end{bmatrix}$$

Pivot

$$\begin{bmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 0 & 2 & -4 & 4 & 2 & -6 \\ 0 & 3 & -6 & 6 & 4 & -5 \end{bmatrix}$$

Pivot

$$\begin{bmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 0 & 2 & -4 & 4 & 2 & -6 \\ 0 & 3 & -6 & 6 & 4 & -5 \end{bmatrix}$$

New pivot column

$$\begin{bmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 0 & 2 & -4 & 4 & 2 & -6 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix}$$

R3 = R3 - 3/2 R2

We've reached Echelon Form

We need to create 0s above each pivot

$$\begin{bmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 0 & 2 & -4 & 4 & 2 & -6 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix}$$

Create 0s here

$$\begin{bmatrix} 3 & -9 & 12 & -9 & 0 & -9 \\ 0 & 2 & -4 & 4 & 0 & -14 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix}$$

R1 = R1 - 6R3
R2 = R2 - 2R3

$$\begin{bmatrix} 3 & -9 & 12 & -9 & 0 & -9 \\ 0 & 1 & -2 & 2 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix}$$

R2 = 1/2 R2

$$\begin{bmatrix} 3 & 0 & -6 & 9 & 0 & -72 \\ 0 & 1 & -2 & 2 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix}$$

R1 = 1/3 R1

$$\begin{bmatrix} 1 & 0 & -2 & 3 & 0 & -24 \\ 0 & 1 & -2 & 2 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix}$$

R1 = R1 + 9R2

We've reached Reduced Echelon Form

Solutions of Linear Systems

$$\begin{bmatrix} 1 & 0 & -5 & 1 \\ 0 & 1 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$



Pivot column Pivot column Not a Pivot column

x_1 x_2 x_3

Basic variables	Free variable
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$$\begin{array}{rcl} x_1 & -5x_3 & = 1 \\ x_2 & + x_3 & = 4 \\ & 0 & = 0 \end{array}$$

Solution:
$$\begin{cases} x_1 = 1 + 5x_3 \\ x_2 = 4 - x_3 \\ x_3 \text{ is free} \end{cases}$$

parametric
representation

Each different choice of x_3 determines a (different) solution of the system, and every solution of the system is determined by a choice of x_3 .

Example

Find the general solution of the linear system whose augmented matrix has been reduced to

$$\begin{bmatrix} 1 & 6 & 2 & -5 & -2 & -4 \\ 0 & 0 & 2 & -8 & -1 & 3 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix}$$

Already in echelon form
 We should reduce to reduced echelon form

$$\begin{array}{l} R1 = R1 + 2R3 \\ R2 = R2 + R3 \end{array} \rightarrow \begin{bmatrix} 1 & 6 & 2 & -5 & 0 & 10 \\ 0 & 0 & 2 & -8 & 0 & 10 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix}$$

$$R2 = 1/2 R2 \rightarrow \begin{bmatrix} 1 & 6 & 2 & -5 & 0 & 10 \\ 0 & 0 & 1 & -4 & 0 & 5 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix}$$

$$R1 = R1 - 2R2 \rightarrow \begin{bmatrix} 1 & 6 & 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & -4 & 0 & 5 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix}$$

$$\begin{array}{rcl} x_1 + 6x_2 + 3x_4 & = & 0 \\ x_3 - 4x_4 & = & 5 \\ x_5 & = & 7 \end{array}$$

The pivot columns are C1, C3, C5
 Therefore, the basic variables are x_1, x_3, x_5
 The free variables are x_2, x_4

Solution:

$$\begin{cases} x_1 = -6x_2 - 3x_4 \\ x_2 \text{ is free} \\ x_3 = 5 + 4x_4 \\ x_4 \text{ is free} \\ x_5 = 7 \end{cases}$$

parametric representation

Parametric representation

We use parametric representation with **consistent** systems that have **infinitely many solutions**

Whenever a system is **inconsistent**, the solution set is **empty**, even when the system has free variables. In this case, the solution set has **no** parametric representation.

Note

Back-Substitution

Consider the following system, whose augmented matrix is in echelon form but is *not* in reduced echelon form:

$$x_1 - 7x_2 + 2x_3 - 5x_4 + 8x_5 = 10$$

$$x_2 - 3x_3 + 3x_4 + x_5 = -5$$

$$x_4 - x_5 = 4$$

A computer program would solve this system by back-substitution, rather than by computing the reduced echelon form. That is, the program would solve **equation 3** for x_4 in terms of x_5 and substitute the expression for x_4 into **equation 2**, solve **equation 2** for x_2 , and then substitute the expressions for x_2 and x_4 into **equation 1** and solve for x_1 .

Our matrix format for the backward phase of row reduction, which produces the reduced echelon form, has the same number of arithmetic operations as back-substitution. But the discipline of the matrix format substantially reduces the likelihood of errors during hand computations. The best strategy is to use only the *reduced* echelon form to solve a system! The *Study Guide* that accompanies this text offers several helpful suggestions for performing row operations accurately and rapidly.

Example

Determine the existence and uniqueness of the solutions to the system

$$\begin{aligned} 3x_2 - 6x_3 + 6x_4 + 4x_5 &= -5 \\ 3x_1 - 7x_2 + 8x_3 - 5x_4 + 8x_5 &= 9 \\ 3x_1 - 9x_2 + 12x_3 - 9x_4 + 6x_5 &= 15 \end{aligned}$$

As we determined in a previous example above,
the echelon form is



$$\begin{bmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 0 & 2 & -4 & 4 & 2 & -6 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix}$$

Is there any false statement ($0 = b$, with b nonzero)? No $\Rightarrow$ The system is consistent

Are there free variables? Yes $\Rightarrow$ the system has infinitely many solutions (solution is not unique)

Theorem 2: Existence and Uniqueness

Theorem

A linear system is consistent if and only if the rightmost column of the augmented matrix is *not* a pivot column—that is, if and only if an echelon form of the augmented matrix has *no* row of the form

$$[0 \ \cdots \ 0 \ b] \quad \text{with } b \text{ nonzero}$$

If a linear system is consistent, then the solution set contains either (i) a unique solution, when there are no free variables, or (ii) infinitely many solutions, when there is at least one free variable.

Using Row Reduction to Solve a Linear System

1. Write the augmented matrix of the system.
2. Use the row reduction algorithm to obtain an equivalent augmented matrix in echelon form. Decide whether the system is consistent. If there is no solution, stop; otherwise, go to the next step.
3. Continue row reduction to obtain the reduced echelon form.
4. Write the system of equations corresponding to the matrix obtained in **step 3**.
5. Rewrite each nonzero equation from **step 4** so that its one basic variable is expressed in terms of any free variables appearing in the equation.

Exercise Find the solution of the following system

$$\begin{array}{rclcl} x_1 - 2x_2 + x_3 & = & 2 & & \\ x_1 - x_2 + 2x_3 & = & 5 & & \\ & x_2 + x_3 & = & 3 & \end{array} \xrightarrow{\text{the echelon form}} \left[\begin{array}{ccc|c} \text{?} & & & \end{array} \right] \xrightarrow{\text{the reduced echelon form}} \left[\begin{array}{ccc|c} \text{?} & & & \end{array} \right]$$

The pivot columns are :

The basic variables are:

The free variables are:

Solution:

?

Check:

Exercise Find the solution of the following system

$$\begin{array}{rclcl} x_1 - 2x_2 + x_3 & = & 2 & & \\ x_1 - x_2 + 2x_3 & = & 5 & & \\ & x_2 + x_3 & = & 3 & \end{array} \xrightarrow{\text{the echelon form}} \begin{bmatrix} 1 & -2 & 1 & 2 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{\text{the reduced echelon form}} \begin{bmatrix} 1 & 0 & 3 & 8 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The pivot columns are :

The basic variables are:

The free variables are:

Solution:
$$\begin{cases} x_1 = 8 - 3x_3 \\ x_2 = 3 - x_3 \\ x_3 \text{ is free} \end{cases}$$

Check:

$$\begin{array}{rclcl} (8 - 3x_3) - 2(3 - x_3) + (x_3) & = & 8 - 3x_3 - 6 + 2x_3 + x_3 & = & 2 \\ (8 - 3x_3) - (3 - x_3) + 2(x_3) & = & 8 - 3x_3 - 3 + x_3 + 2x_3 & = & 5 \\ (3 - x_3) + (x_3) & = & 3 - x_3 + x_3 & = & 3 \end{array}$$

Exercise Find the solution of the following systems

$$\begin{aligned}y + z &= 4, \\ 3x + 6y - 3z &= 3, \\ -2x - 3y + 7z &= 10.\end{aligned}$$

$$\begin{aligned}y + z &= 4, \\ 3x + 6y - 3z &= 3, \\ -2x - 3y + 3z &= 2.\end{aligned}$$

$$\begin{aligned}x + 3y - 2z &= 1, \\ 2x + 6y - 4z &= 2, \\ 3x + 9y - 6z &= 3.\end{aligned}$$

$$\begin{aligned}x_1 + 3x_2 - x_4 &= 0, \\ -3x_1 - 9x_2 + x_3 + x_4 &= 0, \\ 2x_1 + 6x_2 + x_3 - 4x_4 &= 0.\end{aligned}$$

Exercise Find the solution of the following system

$$\begin{aligned}5x - 6y - 7z &= 7 \\6x - 4y + 10z &= -34. \\2x + 4y - 3z &= 29\end{aligned}$$

Exercise Find the solution of the following system

$$\begin{aligned}x + 2y - z &= 3 \\2x - y + 3z &= 5 \\3x + 6y - 3z &= 10\end{aligned}$$